PURAV MAHESH

+65 9039 2435 • e1430679@u.nus.edu • LinkedIn • GitHub

EDUCATION

National University of Singapore Aug 2024 – May 2028 (Expected)
Bachelor of Engineering, Computer Engineering **|** GPA: 4.37 / 5.00
Relevant Coursework: Data Structures & Algorithms, Digital Design, Embedded Systems, Information Security
In Progress: Software Engineering, Machine Learning, Signals and Systems

TECHNICAL SKILLS

Proficient: Python, C/C++, TensorFlow, React, Git
Familiar: Java, Node.js, MongoDB, Prisma, Verilog (HDL)
Domains: Machine Learning, Deep Learning, Edge AI / TinyML, Embedded Systems, RTL / FPGA Design, Full-Stack Web Development

EXPERIENCE

NUS, CS1010E Teaching Assistant Aug 2025 – Present

- Conduct weekly Python tutorials for 25 undergraduates covering control flow, recursion, and algorithmic thinking; hold consultation sessions diagnosing logic errors and improving code quality through structured code review
- Developed supplementary practice problems targeting common student misconceptions, reducing repeat query volume on recurring topics

Raffles Hall Developers (RHDevs), Vice Head Aug 2025 – Present

- Co-led full engineering development of RHApp, a hall management platform for 400+ residents, directing frontend and backend workstreams across a team of student developers
- Organised a NUS-wide hackathon end-to-end — scoping problem statements, coordinating judges and mentors, and managing logistics across multiple competing teams
- Designed and delivered technical workshops on HTML/CSS and Python to 30+ participants, adapted curriculum for secondary school students with no prior programming background

PROJECTS

Emotion Detection & Age Prediction Personal Project, Jan 2024
Python, TensorFlow **|** *github.com/puma-31/EmotionDetection*

- Designed a custom CNN from scratch in TensorFlow for multi-task learning — simultaneously classifying facial emotions across 7 classes and regressing age from a single shared architecture, without pretrained weights
- Achieved 64% emotion classification accuracy on FER2013 (published SOTA ~75% with pretraining); contextualised gap through failure mode analysis across demographic groups, identifying class imbalance as root cause and applying targeted augmentation to improve minority-class recall by reducing false negatives on underrepresented classes
- Built end-to-end ML pipeline: data ingestion, augmentation, architecture iteration, training loop, and quantitative evaluation — documented for reproducibility

Raffles Hall Booking Website Aug 2025
React, JavaScript, Tailwind CSS, MongoDB, Prisma **|** *3-person team*

- Built a full-stack room booking platform used by 100+ hall residents, owning the responsive React/Tailwind frontend and backend authentication APIs with MongoDB and Prisma
- Designed RESTful API endpoints for real-time space availability checks, reducing scheduling conflicts for hall activity spaces; implemented session-based auth with role-differentiated access control

FPGA Scientific Calculator with Graphing Nov 2025
Verilog, Xilinx Basys3 FPGA **|** *3-person team*

- Implemented a fully hardware-based scientific calculator in Verilog on Basys3 FPGA, designing modular RTL components from scratch: ALU, FSM-based control unit, and SPI OLED display driver — no external processor or microcontroller
- Wrote synthesisable Verilog modules with clean state machine design, verified through simulation before on-hardware testing; responsible for SPI display driver integration and timing closure

LinuxLingo — CLI Linux Learning Platform Aug 2025 – Nov 2025

Java, OOP, JUnit | *Team project (CS2113 Software Engineering)* | *github.com/puma-31/tp*

- Contributed to a CLI application that teaches Linux commands interactively in a sandboxed environment; solely responsible for the exam module — designing and implementing question bank management, timed session orchestration, answer validation, and score reporting
- Applied core OOP principles — encapsulation, inheritance, and polymorphism — to structure the exam module with clean separation of concerns across question types, validation logic, and result handling; wrote JUnit tests achieving high coverage of exam-critical code paths
- Practiced rigorous SWE workflows: feature branches, peer code reviews via pull requests, adherence to team coding standards, and incremental delivery aligned to sprint milestones in a 5-person Agile team